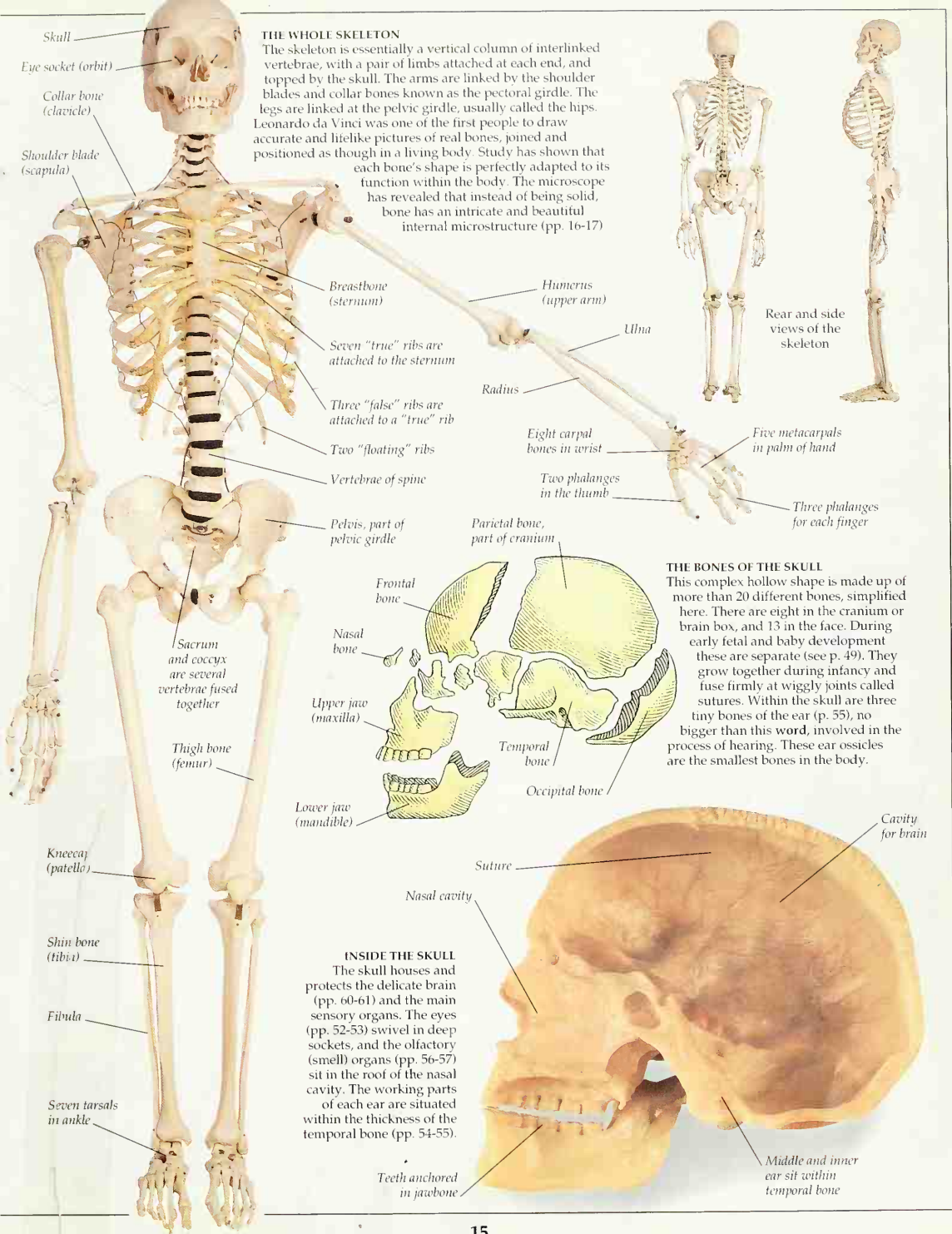




THE WHOLE SKELETON

The skeleton is essentially a vertical column of interlinked vertebrae, with a pair of limbs attached at each end, and topped by the skull. The arms are linked by the shoulder blades and collar bones known as the pectoral girdle. The legs are linked at the pelvic girdle, usually called the hips. Leonardo da Vinci was one of the first people to draw accurate and lifelike pictures of real bones, joined and positioned as though in a living body. Study has shown that each bone's shape is perfectly adapted to its function within the body. The microscope has revealed that instead of being solid, bone has an intricate and beautiful internal microstructure (pp. 16-17)

THE BONES OF THE SKULL

This complex hollow shape is made up of more than 20 different bones, simplified here. There are eight in the cranium or brain box, and 13 in the face. During early fetal and baby development these are separate (see p. 49). They grow together during infancy and fuse firmly at wiggly joints called sutures. Within the skull are three tiny bones of the ear (p. 35), no bigger than this word, involved in the process of hearing. These ear ossicles are the smallest bones in the body.

INSIDE THE SKULL

The skull houses and protects the delicate brain (pp. 60-61) and the main sensory organs. The eyes (pp. 52-53) swivel in deep sockets, and the olfactory (smell) organs (pp. 56-57) sit in the roof of the nasal cavity. The working parts of each ear are situated within the thickness of the temporal bone (pp. 54-55).